

BEYOND STARS: BRIDGING THE GAP BETWEEN RATINGS AND REVIEW SENTIMENT WITH LLMS

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INTRODUCTION

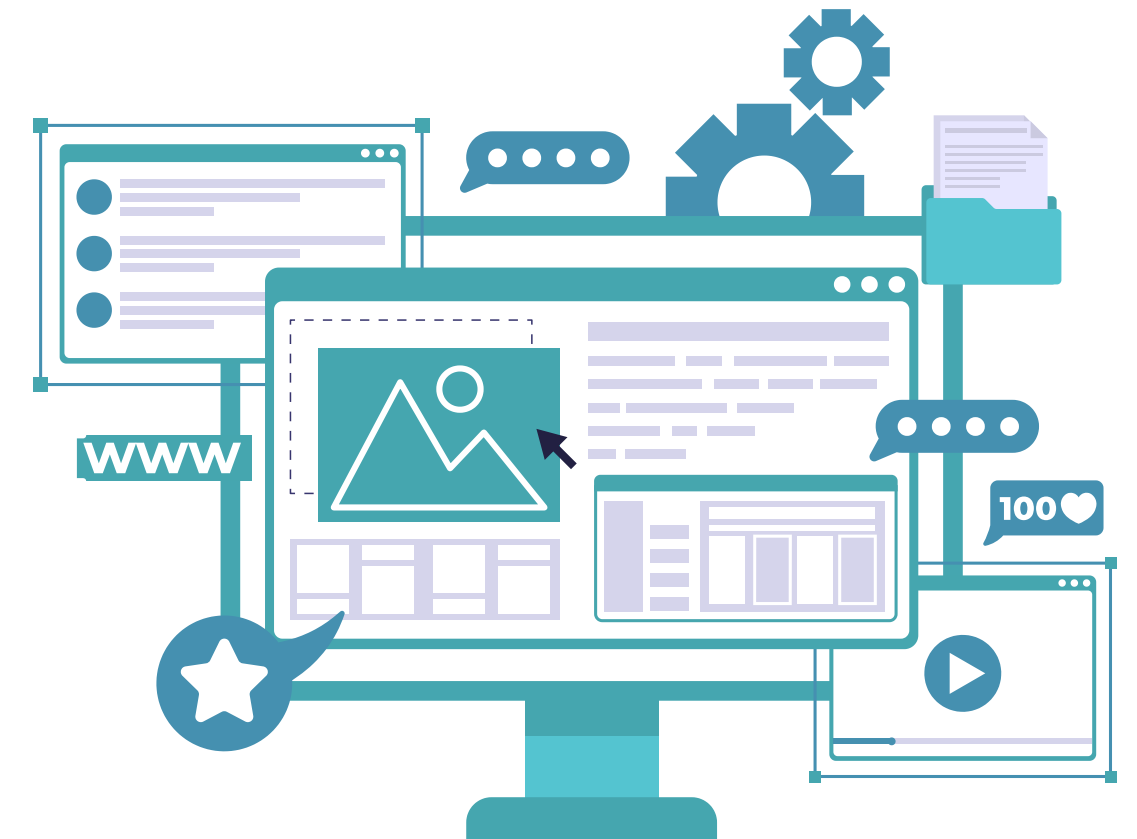
Problem Statement

The Problem

Star ratings often **mismatch the sentiment in reviews** and is inconsistent with the opinion expressed in the review.

The Reason

- Users may give a **star rating** based on their **emotional state**.
- **Reviews** often **focus** on specific **app features** while **star ratings** are meant to **represent an overall impression**.
- Users may **misinterpret the star rating scale**, assigning a score that doesn't align with their written opinion



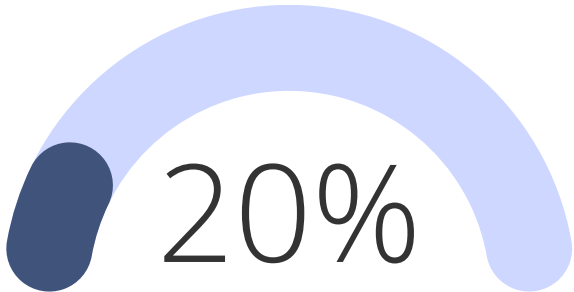
Study Example

Review	App	Actual Rating	Predicted Rating
I love it. I am addicted to it.	Subway Surfers	1	5
App keeps crashing, reinstalled same thing happens FIX IT ASAP	Facebook	5	1
Love it! New favourite social network!	GooglePlus	1	5
I'm 74 and enjoy it. I can't get friends anymore. Good game	AngryBirds	1	4

Table 10: Sample of mismatched reviews across apps (Automatically identified by the DCNN model)

Aralika E, R., IBM Research, Sridhara, G., IBM Research, Gantayat, N., IBM Research, Mani, S., & IBM Research. (2018). Fault in your stars: An Analysis of Android App Reviews [Journal-article].

“Fault in Your Stars: An Analysis of Android App Reviews” by IBM Research

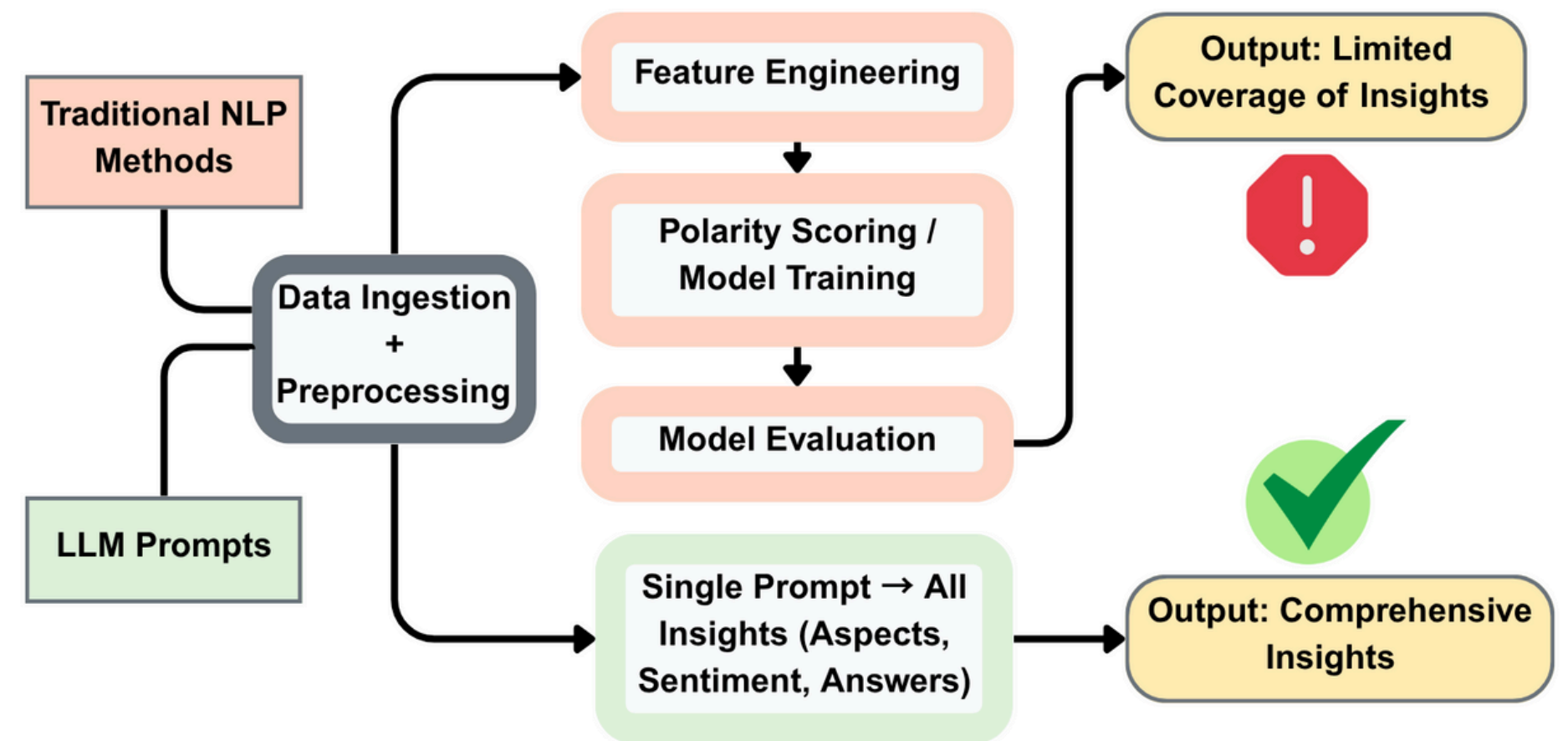


Reviews show discrepancies between ratings and sentiments.



Research Objective

- 1) **Problem baseline:** Quantify **rating-sentiment discrepancies**, addressing the limitation of traditional.
- 2) **Solution contribution:** Develop an **LLM-driven review analysis framework**, providing deeper insight beyond star ratings

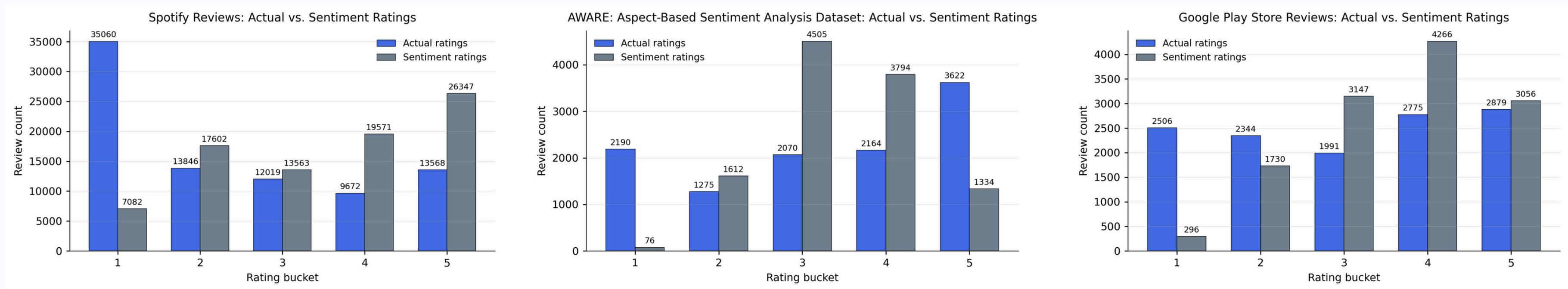


Traditional NLP pipelines vs LLM-based sentiment analysis

METHADODOLOGY

Review Datasets

Discrepancy Analysis: Visualizing the Rating-Sentiment Gap with VADER



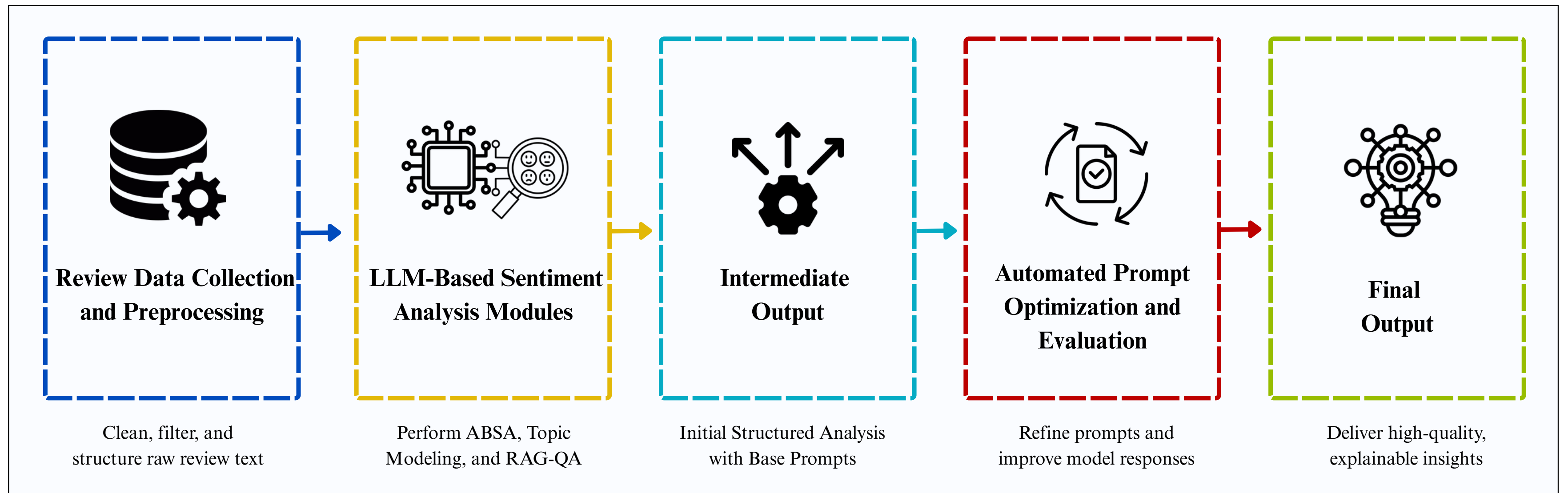
Spotify (80k+)

AWARE (11k+)

Google Play Store (12k+)

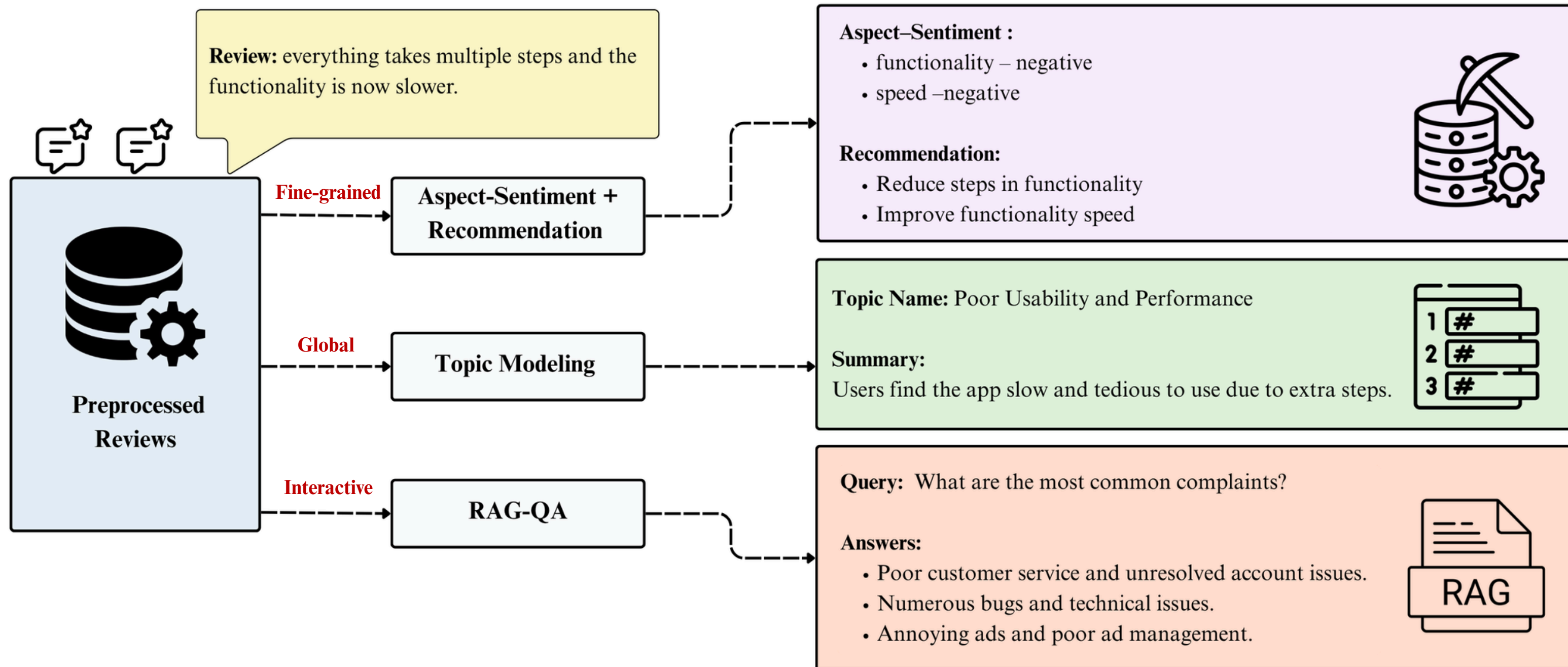
Framework Overview

Proposed Modular Framework for LLM-Based Sentiment Analysis



Main programming language: Python 3.10+

Sample Output



LLM Architecture

Implemented using three diverse LLMs



mistral-small-latest
mistral-large-latest



meta-llama/Llama-2-7b-chat-hf



gpt-4o
gpt-4o-mini
gpt-3.5-turbo

open-source and closed-source LLMs

Prompting Strategy

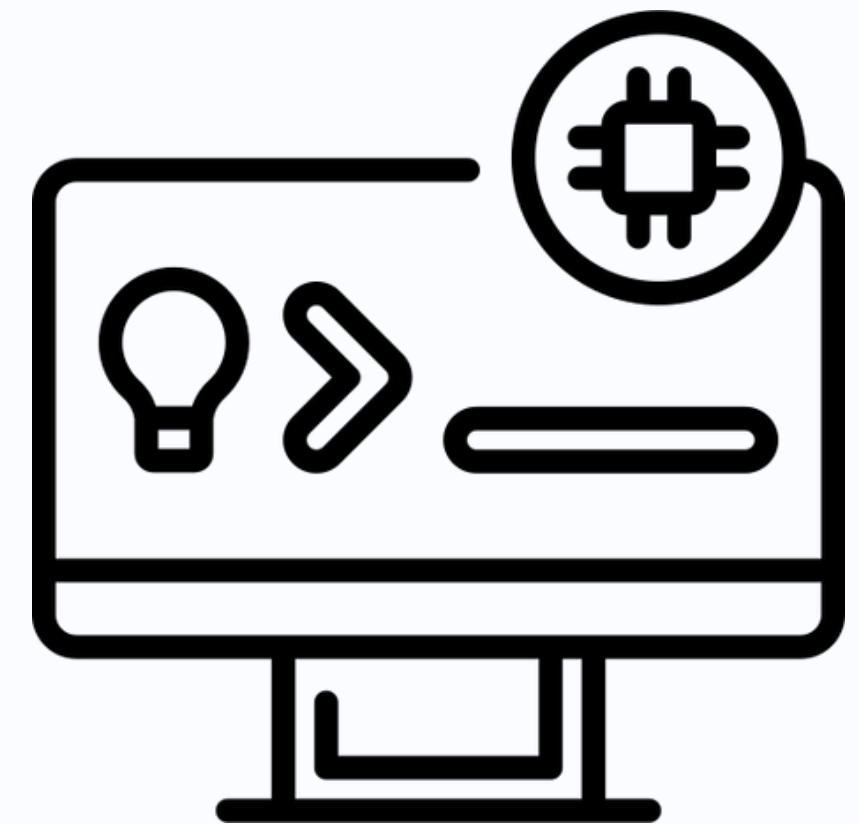
Foundational Prompts Enhanced by Automated Tuning

Stage 1: Build Foundation

- **Zero-Shot** → Direct task instructions
- **Few-Shot** → Examples for complex tasks
- **Chain-of-Thought** → Step-by-step reasoning

Stage 2: Optimize Automatically

- **Meta-Prompting** → Continuous refinement loop
 - a technique where **LLM is used to automatically optimize prompts and evaluate the outputs** of those prompts



RESULTS OF EXPERIMENTS

Aspect Extraction with Recommendation Mining

TABLE V
ASPECT EXTRACTION PERFORMANCE ON AWARE DATASET

Model	Precision	Recall	F1-Score
GPT-4 (Prompt-based)	0.892	0.892	0.892
DeBERTa-v3-large	0.847	0.835	0.841

TABLE VI
PREDICTED SENTIMENT DISTRIBUTION ON AWARE DATASET

Model	Positive	Negative	Neutral
LLM-based	22.3%	29.8%	47.9%
VADER	8.7%	3.1%	88.2%

→ Uses AWARE's **human annotated** aspect terms and categories **as ground truth**.

→ **+5.1 % improvement in F1**; better handling of implicit features.

→ Include triples (**aspect, sentiment, recommendation**) .

→ LLM-based model generates a **balanced sentiment distribution**, overcoming the strong neutral bias shown by VADER.

Topic Modeling

TABLE VIII
SILHOUETTE COEFFICIENTS FOR TOPIC MODELING
ON SPOTIFY DATASET

Metric	BERTopic Only	BERTopic + LLM
Silhouette Score	-0.0313	0.0302

Silhouette Score: A metric that measures how similar a data point is to its own cluster compared to other clusters.

→ Assess the improvement gained by **integrating an LLM into the BERTopic pipeline** on the Spotify dataset.

→ LLM-enhanced approach **improves silhouette from negative to positive.**

→ Indicating **enhanced cluster separation**, topic distinctiveness, and validating the benefit of LLM generated labels for **interpretability.**

Retrieval-augmented question answering (RAG-QA)

TABLE IX
RETRIEVAL PROXY METRICS (K=10) FOR SELECTED SPOTIFY QUERIES
(HIGHER DIVERSITY IS BETTER)

Query	Avg. Cosine Sim.	Diversity
What complaints do users have about Spotify's offline-mode buffering?	0.713	1.0
What do listeners say about Spotify crashing or freezing on startup?	0.754	1.0
How do listeners describe the app's offline playback experience?	0.696	1.0
How do users report errors or failures when downloading songs for offline use?	0.618	1.0
What do users say about Spotify's crossfade and track-transition experience?	0.650	1.0

Cosine Similarity: Measures how closely the search results match the question.

Diversity: Measures whether the answers come from many different reviews.

→ **Tested on 5** targeted Spotify **queries**.

→ Achieved **perfect diversity and high cosine similarity**; both highly relevant and drawn from a wide range of user reviews.

→ **Human validation** confirmed the **LLM's response** were accurate, as they correctly **reflected the content of the top 10 review chunks** retrieved for each query.

CONCLUSION AND FUTURE WORK

Conclusion: Key Takeaways

- **Problem:** Star ratings are often misleading and don't reflect what users write.
- **Our Solution:** A new AI system that deeply analyzes reviews using LLMs.
- **Key Innovations:** Modular framework, Multi-level insights, Automated prompt refinement, Multi-LLM backends.
- **Experiments Result:** Outperforms traditional methods, providing more accurate and useful insights.
- **Impact:** Helps developers build better apps by truly understanding user needs.

Future Work: Next Steps

1. Framework Enhancement & Generalization

- Extend to **Multilingual & Cross-Domain Scenarios**
- Develop **Agentic Architectures** that can **dynamically adapt prompts** in real-time based on feedback and new data.

2. Robustness & Comprehensive Evaluation

- **Test the system's reliability** through **adversarial testing**.
- Conduct **Detailed Error Analysis** on prompt failures and hallucinations
- **Benchmark LLM Architectures** through comparative analysis of different LLMs to understand **performance-cost trade-offs**.
- Perform **statistical significance testing** to validate observed improvements and ensure the robustness of experimental results.



THANK YOU!

Open for Questions.....

For more Information



Github Repo



Paper